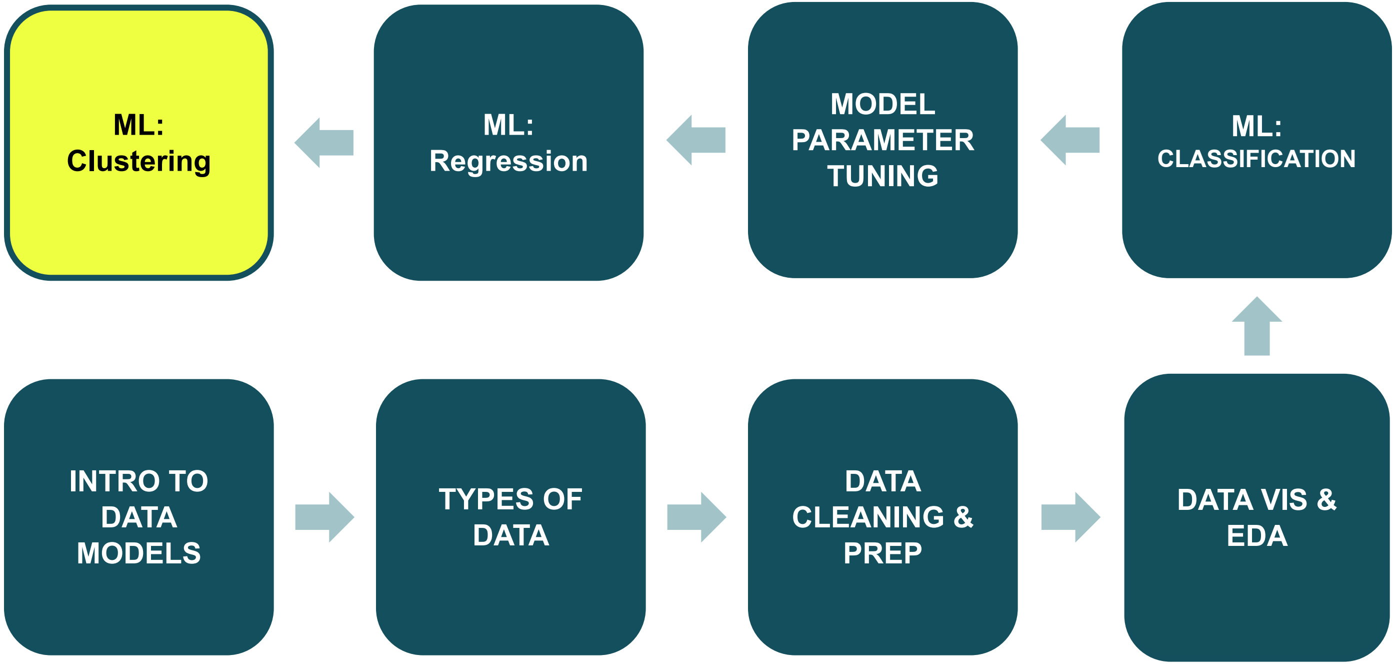
ML: Clustering

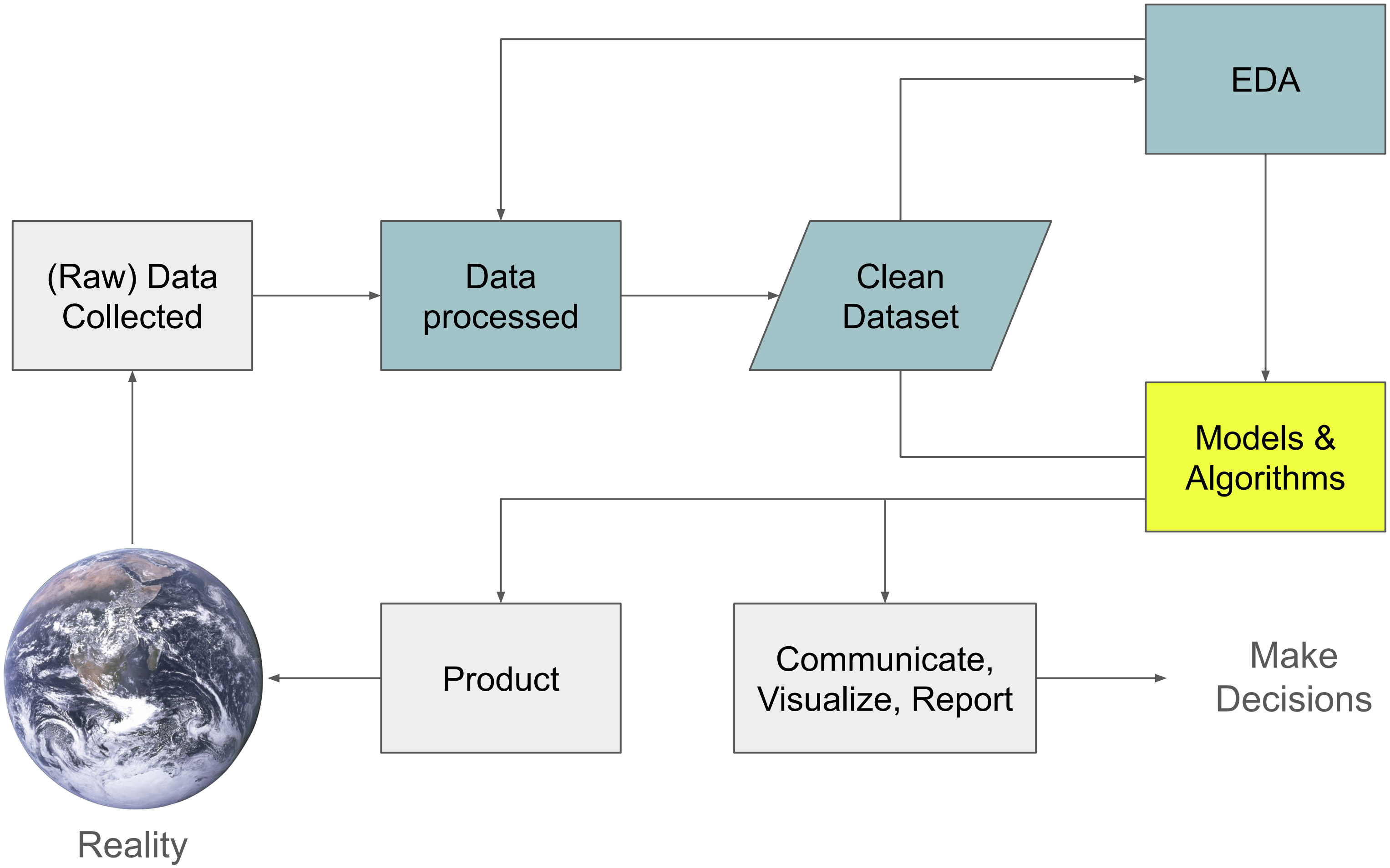
CPE 232: Data Models

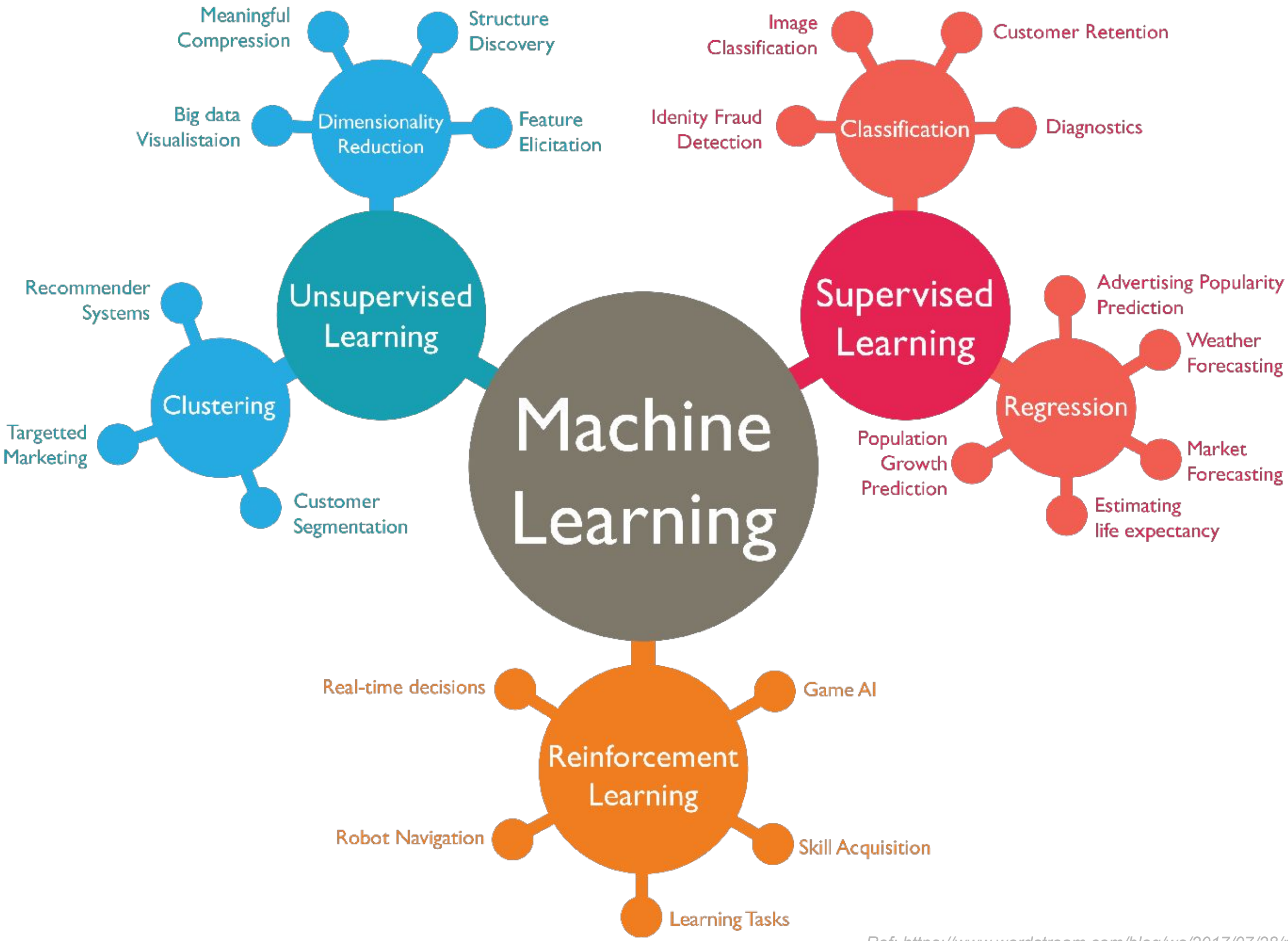
Dr. Sansiri Tarnpradab

Department of Computer Engineering, KMUTT

Review


ML:
Clustering
ML:
Regression
MODEL
PARAMETER
TUNING
ML:
CLASSIFICATION
INTRO TO
DATA
MODELS
TYPES OF
DATA
DATA
CLEANING &
PREP
DATA VIS &
EDA


EDA
(Raw) Data Collected
Data processed
Clean Dataset
Models & Algorithms
Product
Communicate, Visualize, Report
Make Decisions
Reality



Ref: https://www.wordstream.com/blog/ws/2017/07/28/machine-learning-applications

Basic Types of Machine Learning

Labeled Data

Example inputs
Example outputs

Supervised Learning Algorithms

Unseen Inputs

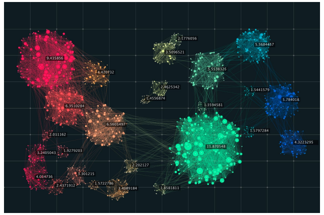
Prediction Model
(Classification/Regression)

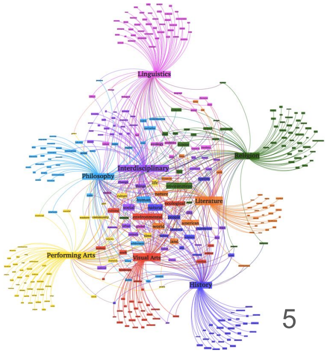
Predicted Outputs

Unlabeled Input Data

Unsupervised Learning Algorithms

Hidden/Unusual patterns
Alternative representations



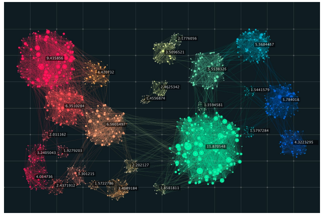


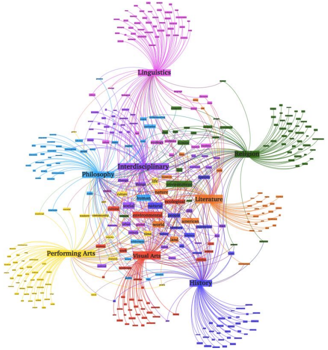
Refs:
[Face Detection]
[Spam Detection]
[Topic Modeling]
[Social Networks]

Outline

- Unsupervised-learning
- Revisiting Data Points
- Clustering Concept
  - Similarity Measures
  - Distance Functions
  - Quality of Clustering
- Clustering Approaches
  - Partitioning-based
  - Hierarchical-based
  - Density-based

Unsupervised-learning





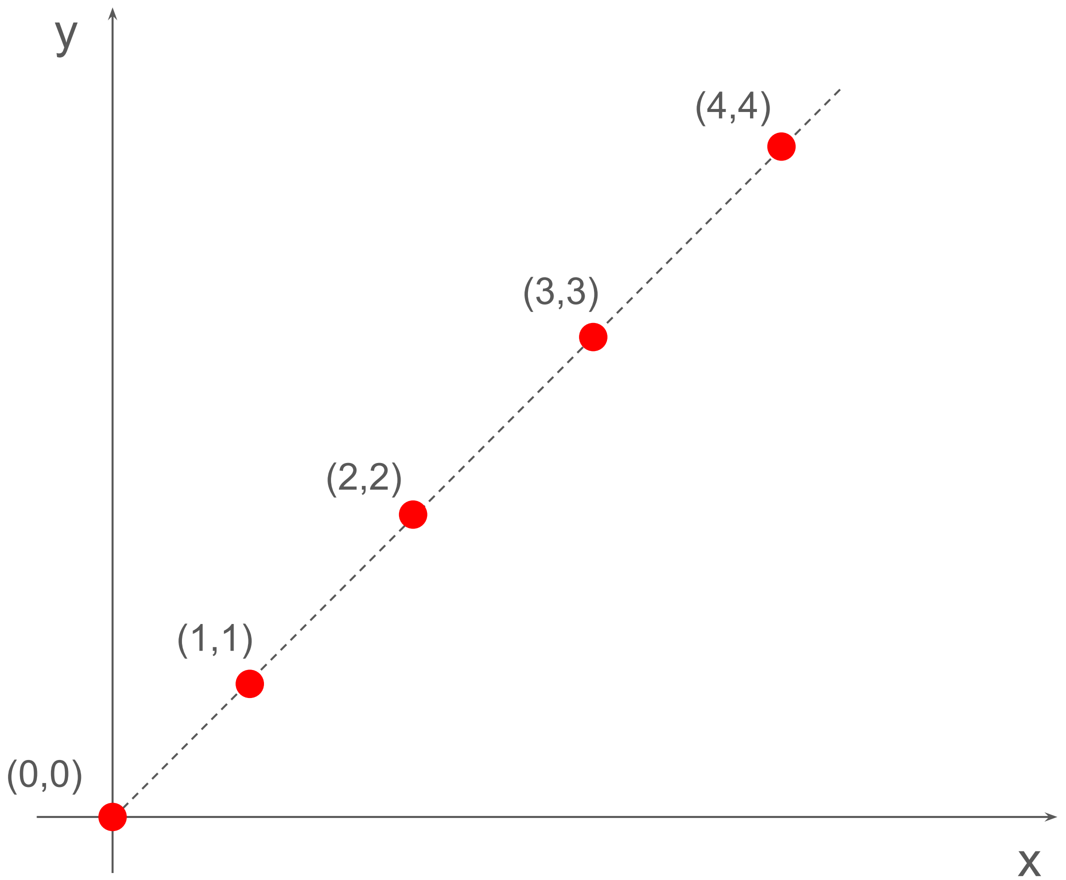
Unlabeled Input Data → **Unsupervised Learning** Algorithms → Hidden/Unusual patterns Alternative representations

- No predefined classes
- No labels needed
- Applications:
    - Stand-alone tool to get insight into data distribution
    - A preprocessing step for other algorithms

A ***data point*** represents a single piece of information.

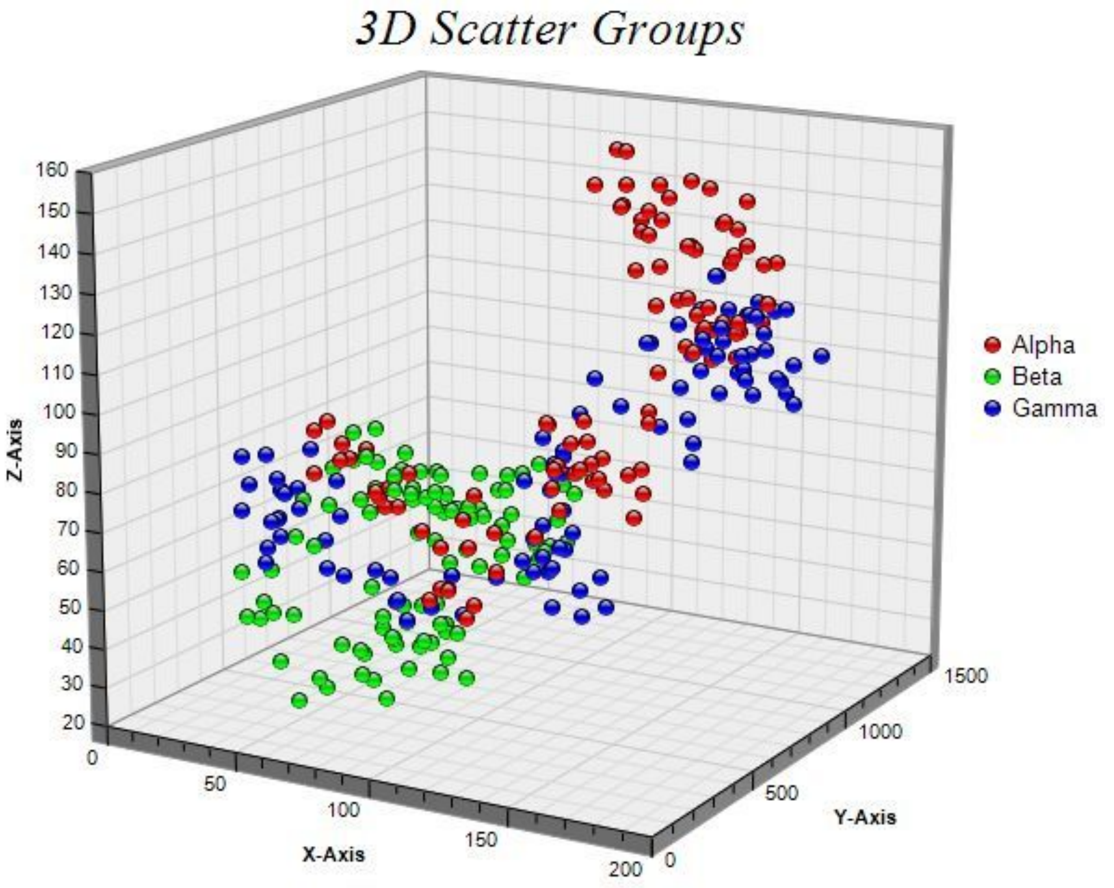
Data Points
2D

x	y
0	0
1	1
2	2
3	3
4	4



Data Points
3D

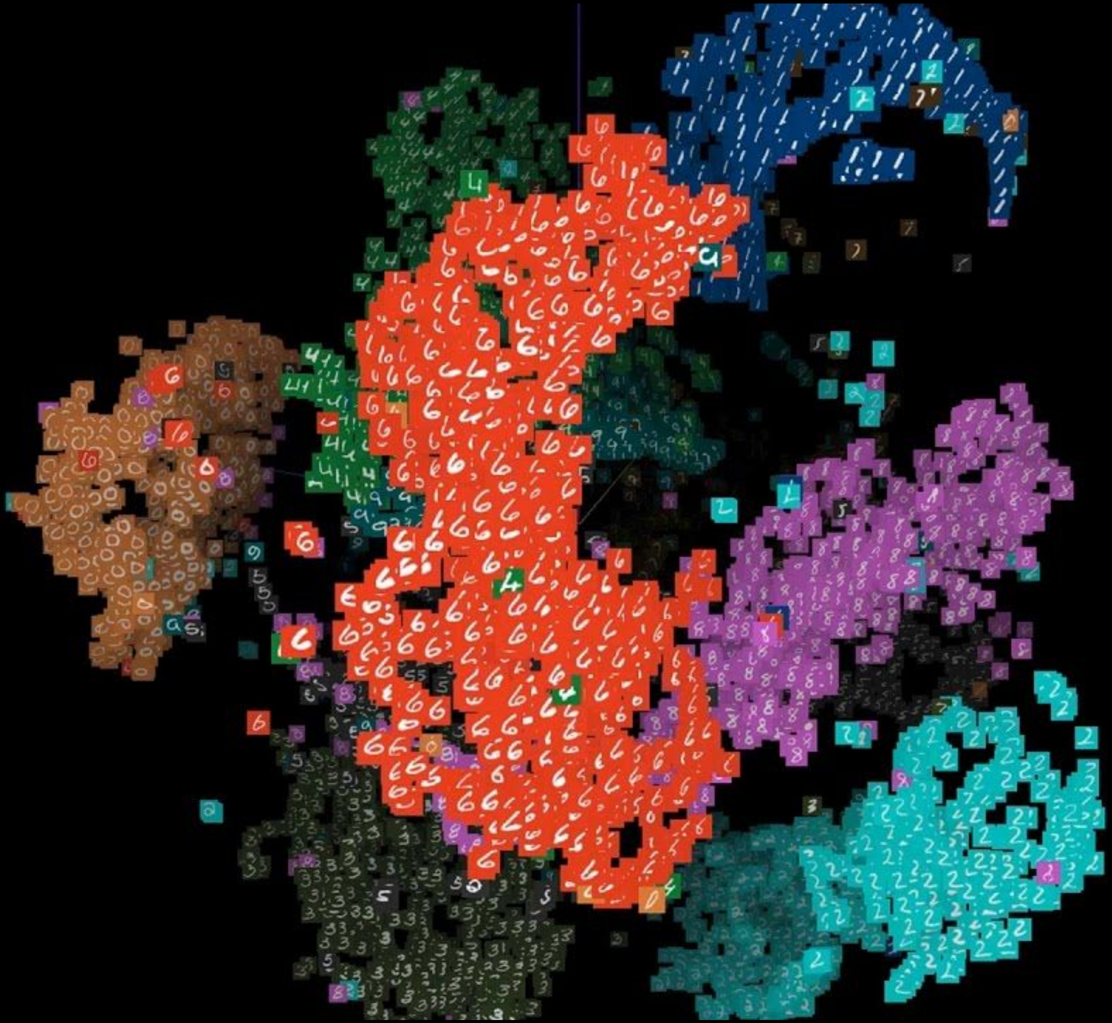
x	y	z
?	?	?
?	?	?
?	?	?
?	?	?
?	?	?



Ref: https://forum.posit.co/t/3d-scatter-plot-with-data-points-colored-according-to-their-group/99145

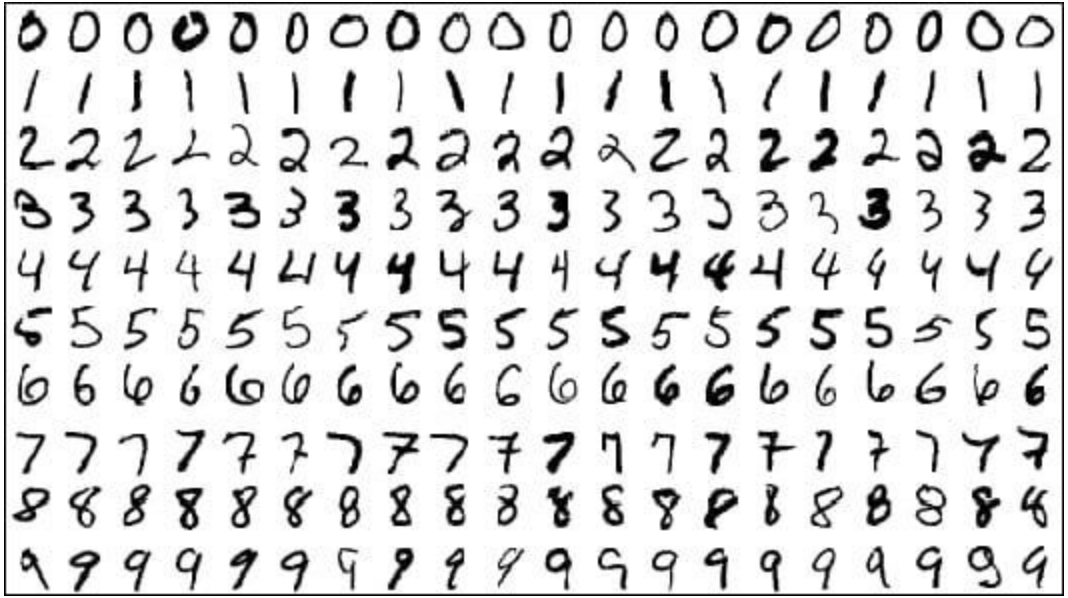
Data Points
High-dimensional Space

Attribute 1	Attribute 2	…	Attribute k
x_{11}	x_{12}	…	x_{1k}
x_{21}	x_{22}	…	x_{2k}
x_{31}	x_{32}	…	x_{3k}
…	…	…	…
x_{n1}	x_{n2}	…	x_{nk}



Ref: https://www.youtube.com/watch?v=wvsE8jm1GzE

MNIST 0-9



Ref: https://medium.com/@navyashree.raghupatro/recognizing-handwritten-digits-with-scikit-learn-8d248dc01b6d

Cluster of Data Points

Cluster **Analysis**

Cluster

A collection of data objects

Similarity

Dissimilarity

Cluster Analysis

Finding Similarities

Characteristics

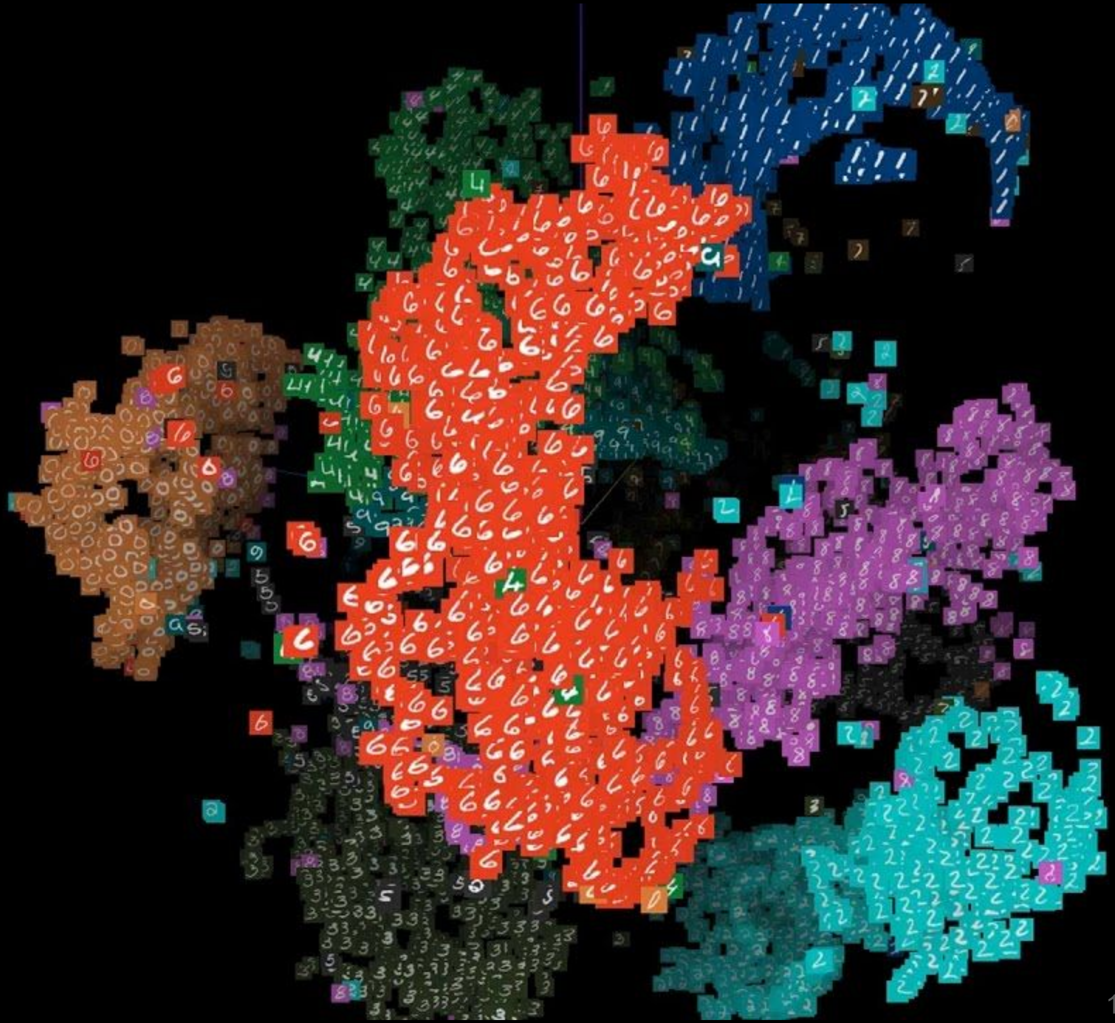
Grouping based on similarities

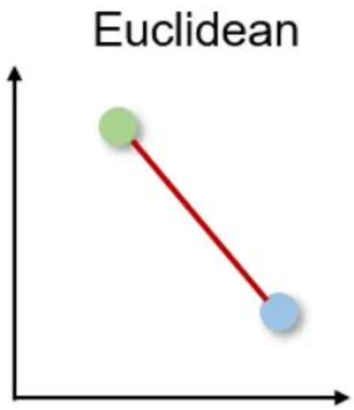
Similarity

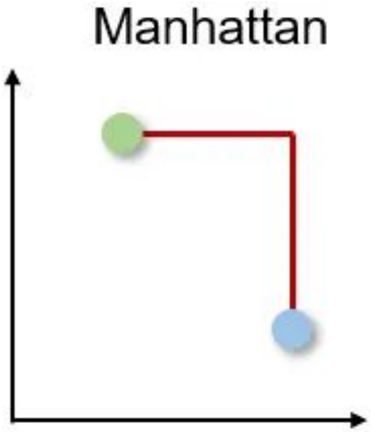
- If two things are similar in some ways, they often share other characteristics

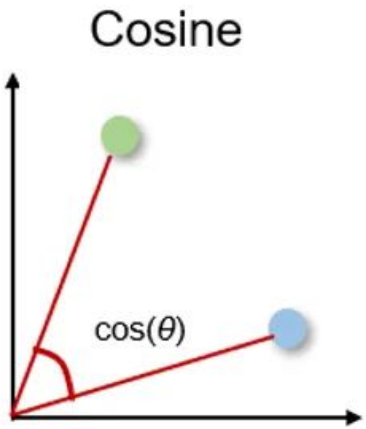
- Applications:
  - Recommendations (e.g. Netflix, Amazon)
  - Troubleshooting
  - Knowledge management
  - Customer segmentation
  - Even classification or regression

Determine Similarities







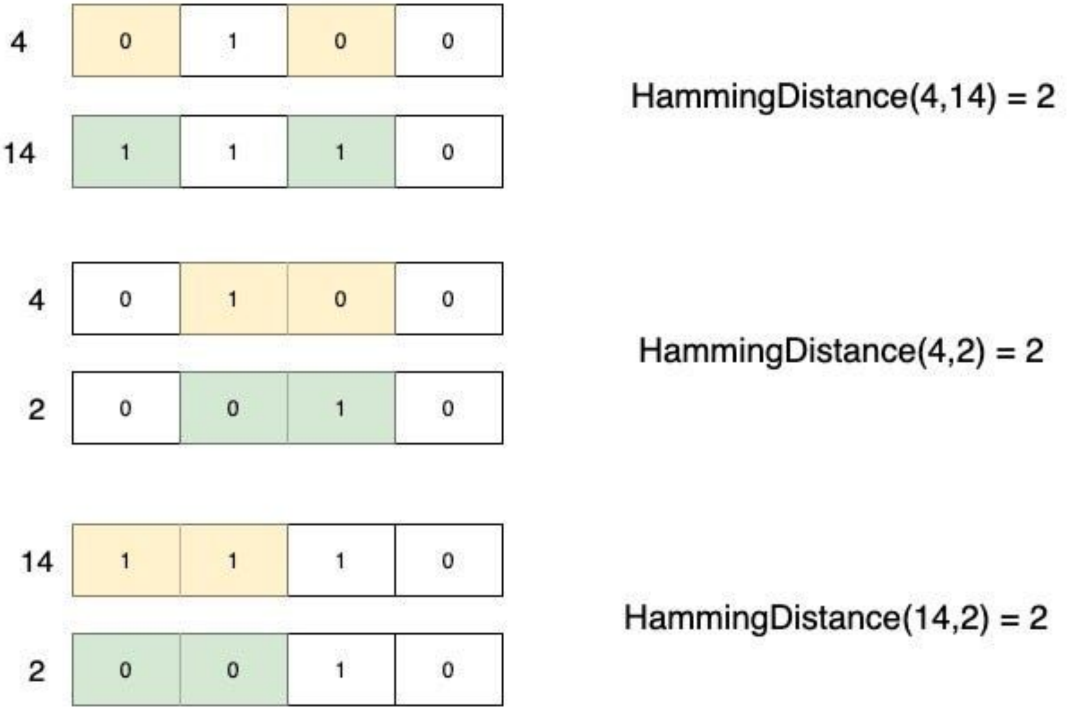


- Shortest distance between two real-valued vectors
- Most common

- Taxicab or City-block distance
- Shortest distance between two real-valued vectors
- Right angles

- Cosine between two vectors
- Often used in higher dimensionality
- Measured in Θ
  - Θ = 0° → similar (overlap)
  - Θ = 90° → dissimilar

Ref: https://medium.com/@jodancker/a-brief-introduction-to-distance-measures-ac89cbd2298



Ref: https://medium.com/geekculture/total-hamming-distance-problem-1b74decd71c9

- Number of positions where two vectors (or strings) differ
- Used for binary or categorical data
- Same length required
- Examples: Compare…
  - Student answer sheets
  - Passwords
  - DNA sequences

Distance Functions

Euclidean Distance

$$d_{Euclidean}(X,Y) = ||X-Y||_2 = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + \ldots}$$

Manhattan Distance

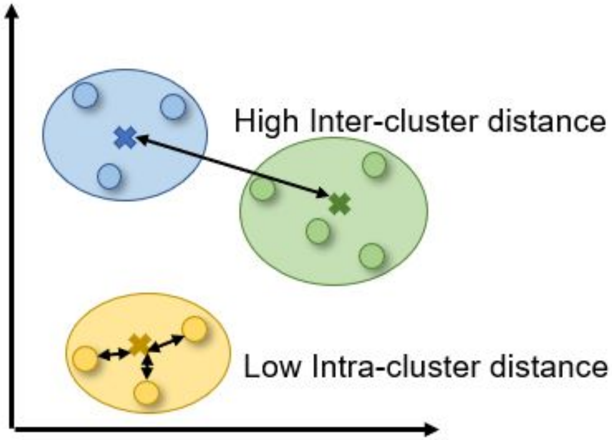
$$d_{Manhattan}(X,Y) = ||X-Y||_1 = |x_1-y_1| + |x_2-y_2| + \ldots$$

Cosine Distance

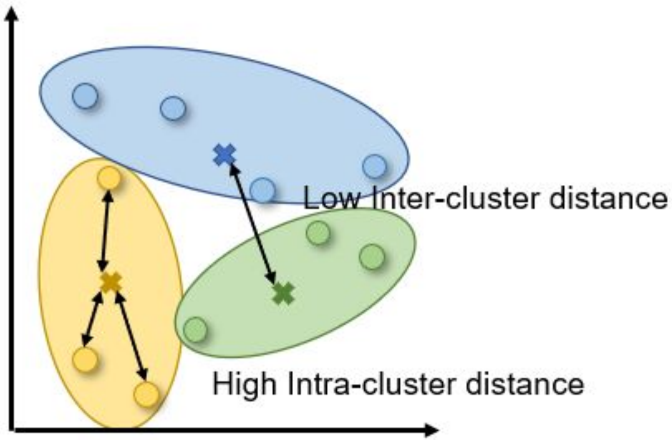
$$d_{Cosine}(X,Y) = \frac{X \cdot Y}{||X|| \times ||Y||}$$

Quality of Clustering

- High *intra*-class similarity
- Low *inter*-class similarity



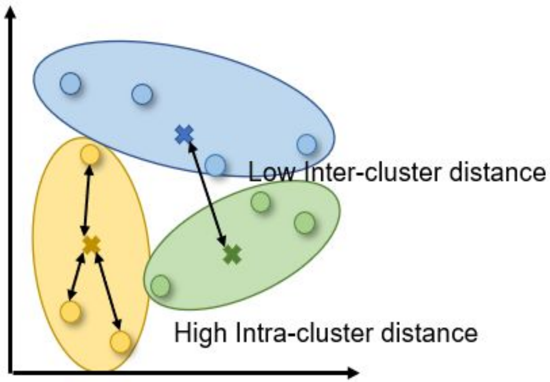
Good Example

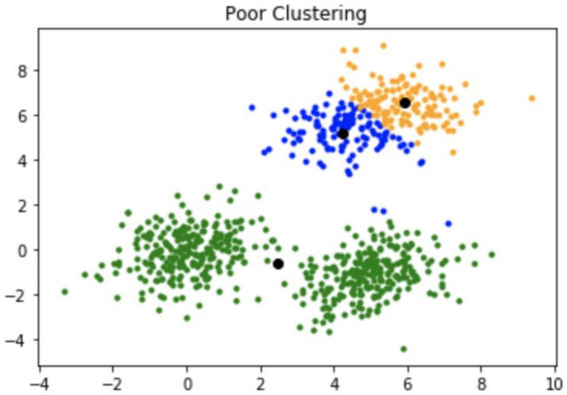


Bad Example

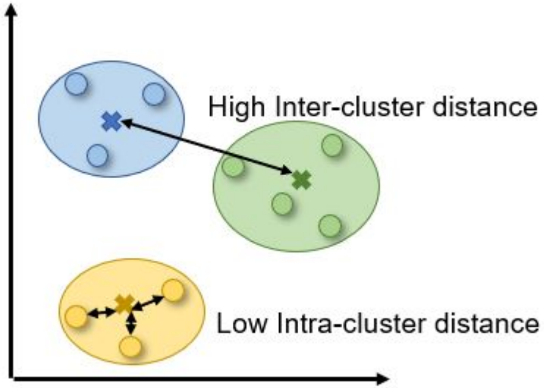
Ref: https://medium.com/@jodancker/a-brief-introduction-to-cluster-validation-ca4215295b06

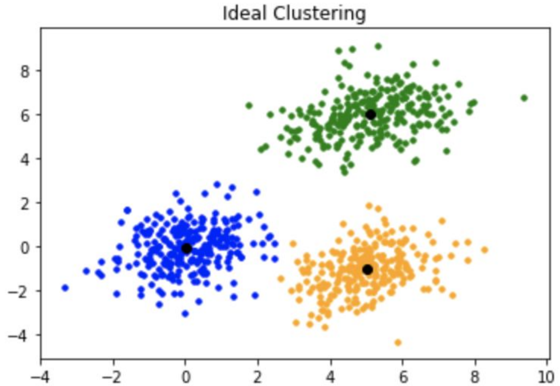
Bad Example





Good Example





Refs:
https://www.geeksforgeeks.org/ml-k-means-algorithm/
https://medium.com/@jodancker/a-brief-introduction-to-cluster-validation-ca4215295b06

Clustering Approaches

- **Partitioning** Approach
- **Hierarchical** Approach
- **Density-based** Approach

Clustering Approaches

- **Partitioning** Approach
- **Hierarchical** Approach
- **Density-based** Approach

Partitioning: Basic Concept

- Breaking down a large group of data points into partitions
- While still taking into account the distance → minimum

Basic Concept

Construct a partition of a database D of n objects into a set of k clusters, such that sum of squared distance is minimal

Partitioning: Brute-force

Finding a global optimal clustering:

1. Exhaustively enumerate <u>ALL</u> the clusterings

2. Return the clustering with the min score

$$\hat{C} = \arg\min_{C} \{SSE(C)\}$$

Basic Concept

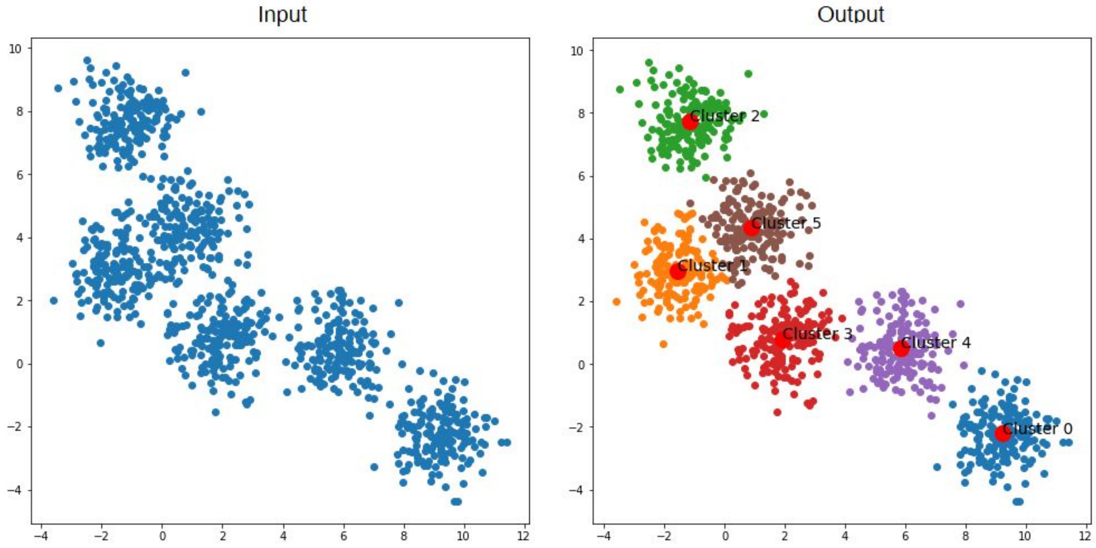
Construct a partition of a database D of n objects into a set of k clusters, such that sum of squared distance is minimal

k=2 k=3 k=4 k=5 k=...

Computationally Infeasible

Partitioning: K-means

- Each cluster is represented by the center of the cluster
- **Centroid** → Center of the cluster → Average



Ref: https://towardsai.net/p/l/centroid-neural-network-an-efficient-and-stable-clustering-algorithm

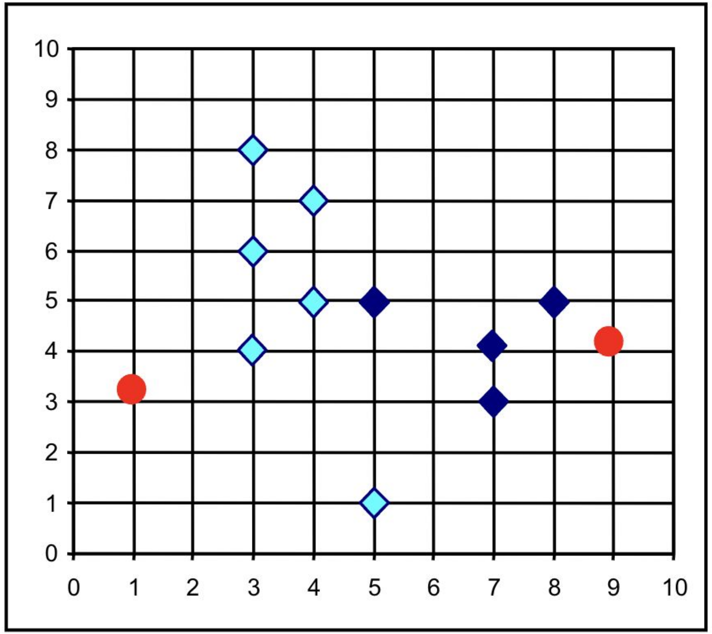
k-means Steps

1. Partition objects into k non-empty subsets.
2. Compute seed points as the centroids of the clusters of the current partition.
3. Assign each object to the cluster with the nearest seed point.
4. Go back to Step 2, stop when no more new assignment.

k-means Steps (1-2)

1. Partition objects into k non-empty subsets. (k=2)

2. Compute seed points as the centroids of the clusters of the current partition.

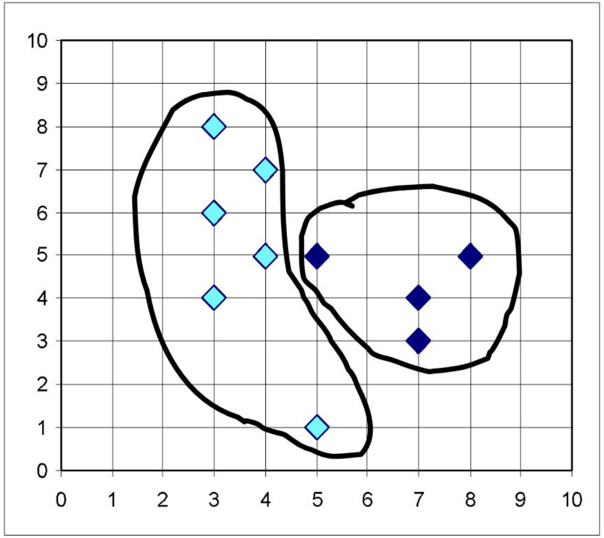


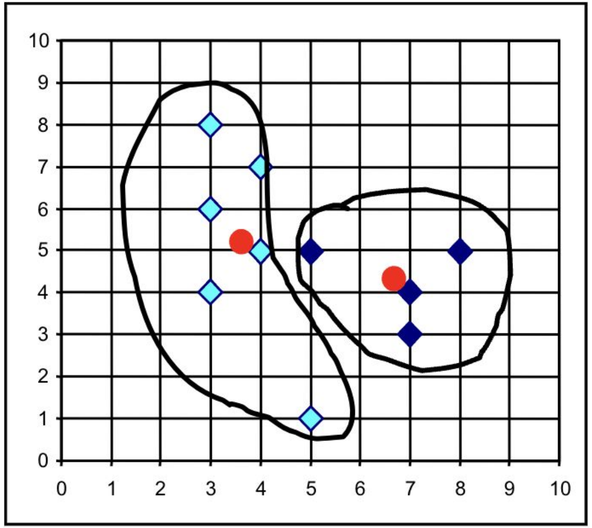
k-means Steps (3-4)

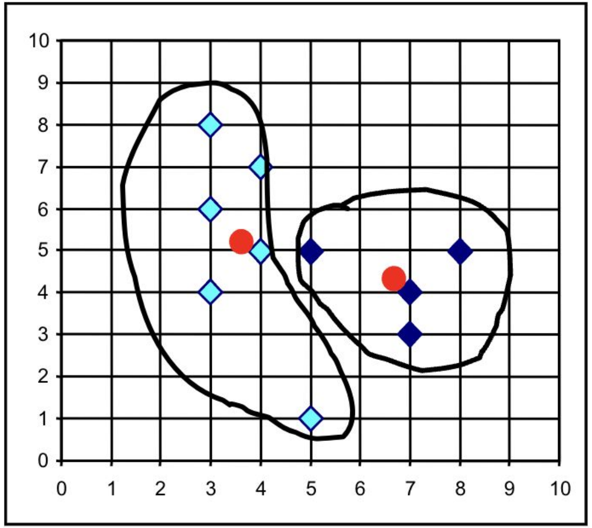
Step 2: Compute seed points as the centroids of the clusters of the current partition.

3. Assign each object to the cluster with the nearest seed point.

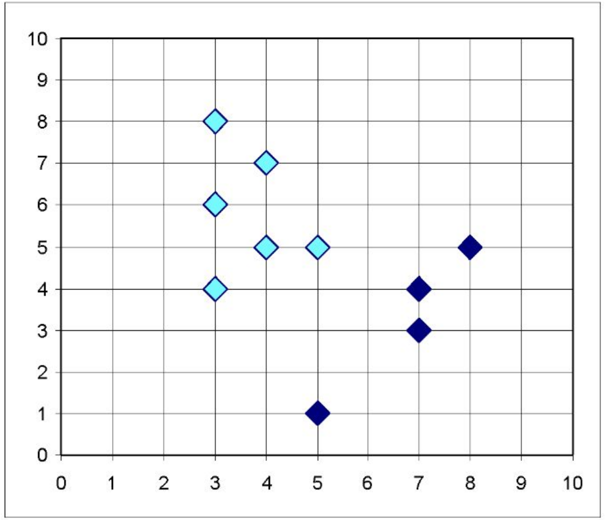
4. Go back to **Step 2**, stop when no more new assignment.



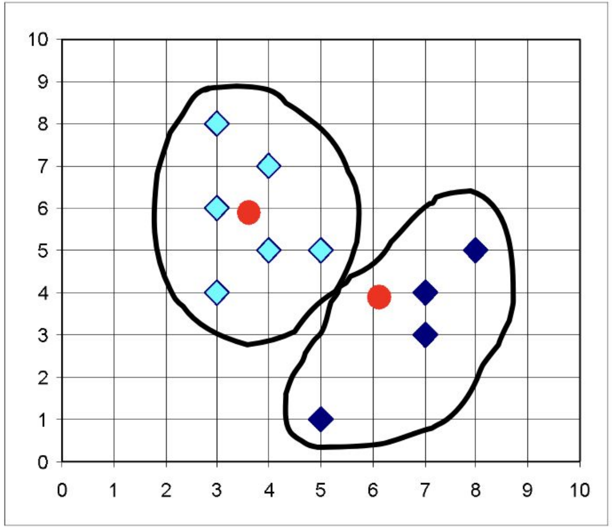




10
9
8
7
6
5
4
3
2
1
0
0 1 2 3 4 5 6 7 8 9 10



10
9
8
7
6
5
4
3
2
1
0
0 1 2 3 4 5 6 7 8 9 10

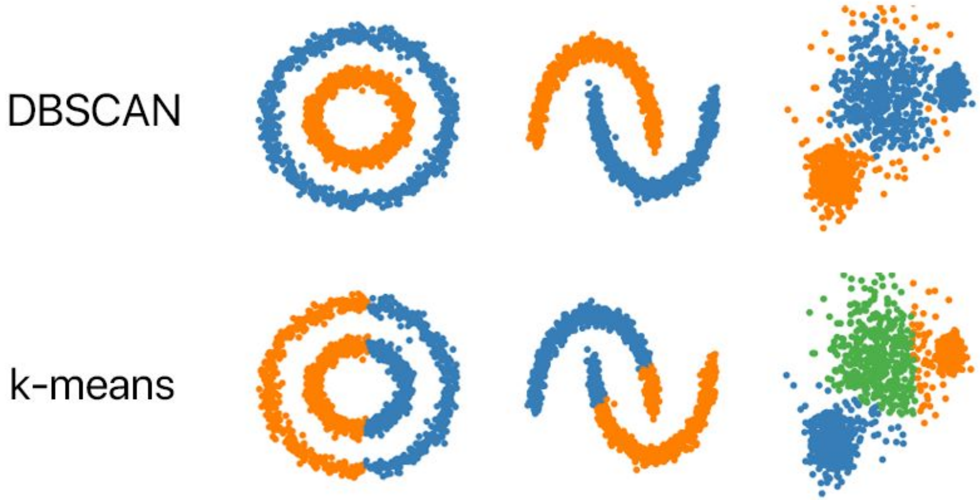


10
9
8
7
6
5
4
3
2
1
0
0 1 2 3 4 5 6 7 8 9 10

Things to consider…

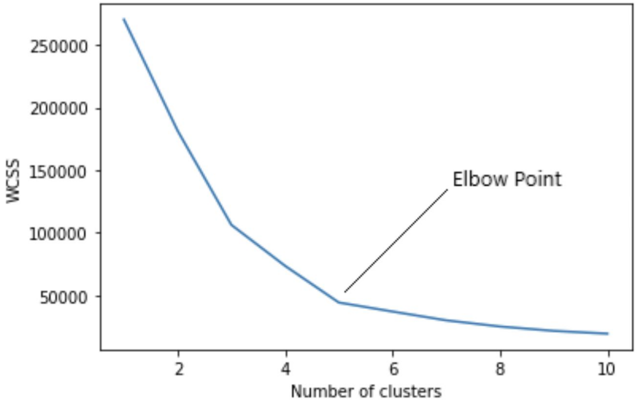
- What should be the value of k?
- Methods to determine k
    - **Elbow**: Plot the within-cluster sum of squares (WCSS) against the number of clusters.
    - **Silhouette Score**: Measures how similar an object is to its own cluster vs. others.
        - Values range from -1 to 1
        - +1 → Perfectly clustered.
        - 0 → The point is on or very close to the decision boundary between two clusters.
        - -1 → The point is likely assigned to the wrong cluster.
- Outlier matters?
    - k-medoids (uses medians rather than means).
    - DBSCAN (density-based; automatically handles noise/outliers).

k-Means VS DBSCAN



Ref: https://github.com/NSHipster/DBSCAN

Elbow **Method**



- Plot output from different k values
- Identify the elbow point
- WCSS *(Within-Cluster Sum of Square)*
    - Squared average distance of all the points within a cluster *to the cluster centroid*
    - As the number of clusters increases, the WCSS value decreases
- Weaknesses
    - Only works for k-means clustering
    - May not hold for complex datasets with irregularly shaped or differently sized clusters
    - Sensitive to initial cluster centroids
    - Inefficient for large datasets

Ref elbow image:
https://excelsportspt.com/two-exercises-help-pain-when-elbow-is-bent-or-straightened/

WCSS Example

A	(1, 2)
B	(1, 4)
C	(1, 0)
D	(10, 2)
E	(10, 4)
F	(10, 0)

Let's assume k = 2

- Cluster 1: {A, B, C}
- Cluster 2: {D, E, F}

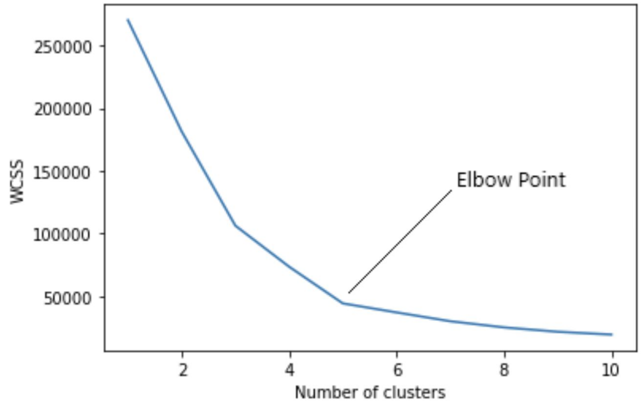
Centroid for each cluster…

- Cluster 1: ((1+1+1)/3, (2+4+0)/3) → (1, 2)
- Cluster 2: ((10+10+10)/3, (2+4+0)/3) → (10, 2)

Compute squared distance to centroid *(via Euclidean)*:

- Cluster 1: A(0), B(2^2), C(2^2) → $WCSS_1$ = 8
- Cluster 2: D(0), E(2^2), F(2^2) → $WCSS_2$ = 8

$WCSS_{Total}$ = 16

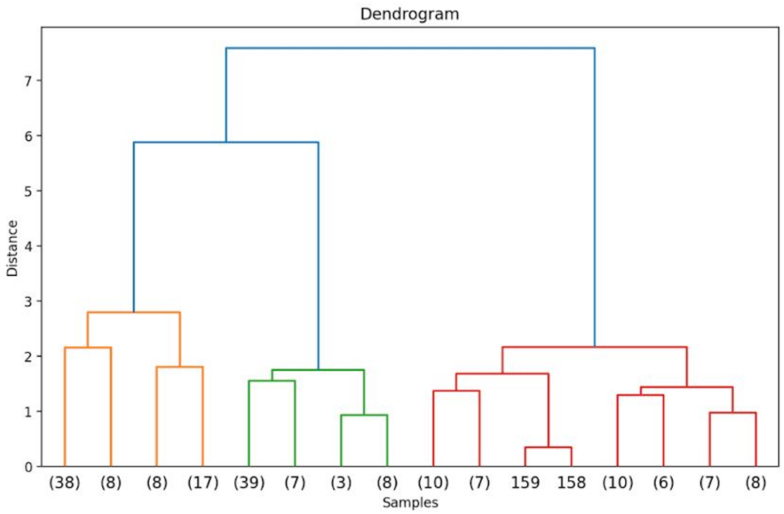


Clustering Approaches

- **Partitioning** Approach
- **Hierarchical** Approach
- **Density-based** Approach

Hierarchical Methods

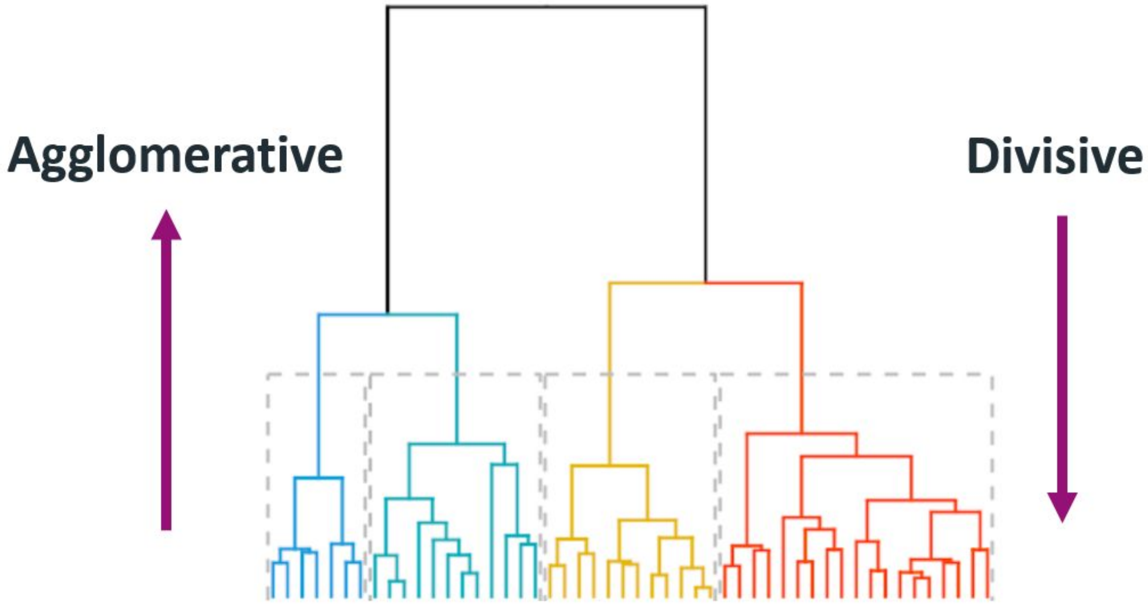
- Produces a set of nested clusters
- Organized as a hierarchical tree



Ref: https://www.kdnuggets.com/unveiling-hidden-patterns-an-introduction-to-hierarchical-clustering

Hierarchical: Basic Concept

- Merge or split one cluster at a time
- Merge → Agglomerative
- Split → Divisive



Ref: https://harshsharma1091996.medium.com/hierarchical-clustering-996745fe656b

Hierarchical: Agglomerative

Compute the proximity matrix

Let each data point be a cluster

Repeat

Merge the two closest clusters

Update the proximity matrix

Until only a single cluster remains

	P1	P2	P3	P4	P5
P1					
P2					
P3					
P4					
P5					

Proximity Matrix

Hierarchical: Agglomerative Example

P1	(0.4, 0.53)
P2	(0.22, 0.38)
P3	(0.35, 0.32)
P4	(0.26, 0.19)
P5	(0.08, 0.41)
P6	(0.45, 0.30)

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.14	0			
P4	0.37	0.19	0.16	0		
P5	0.34	0.14	0.28	0.28	0	
P6	0.24	0.24	0.10	0.22	0.39	0

P1	(0.4, 0.53)
P2	(0.22, 0.38)
P3	(0.35, 0.32)
P4	(0.26, 0.19)
P5	(0.08, 0.41)
P6	(0.45, 0.30)

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.14	0			
P4	0.37	0.19	0.16	0		
P5	0.34	0.14	0.28	0.28	0	
P6	0.24	0.24	**0.10**	0.22	0.39	0

According to the proximity matrix, a cluster can be formed by merging points that are closest to each other – **(P3, P6) = 0.10**

P3 P6

	P1	P2	P3, P6	P4	P5
P1	0				
P2		0			
P3, P6			0		
P4				0	
P5					0

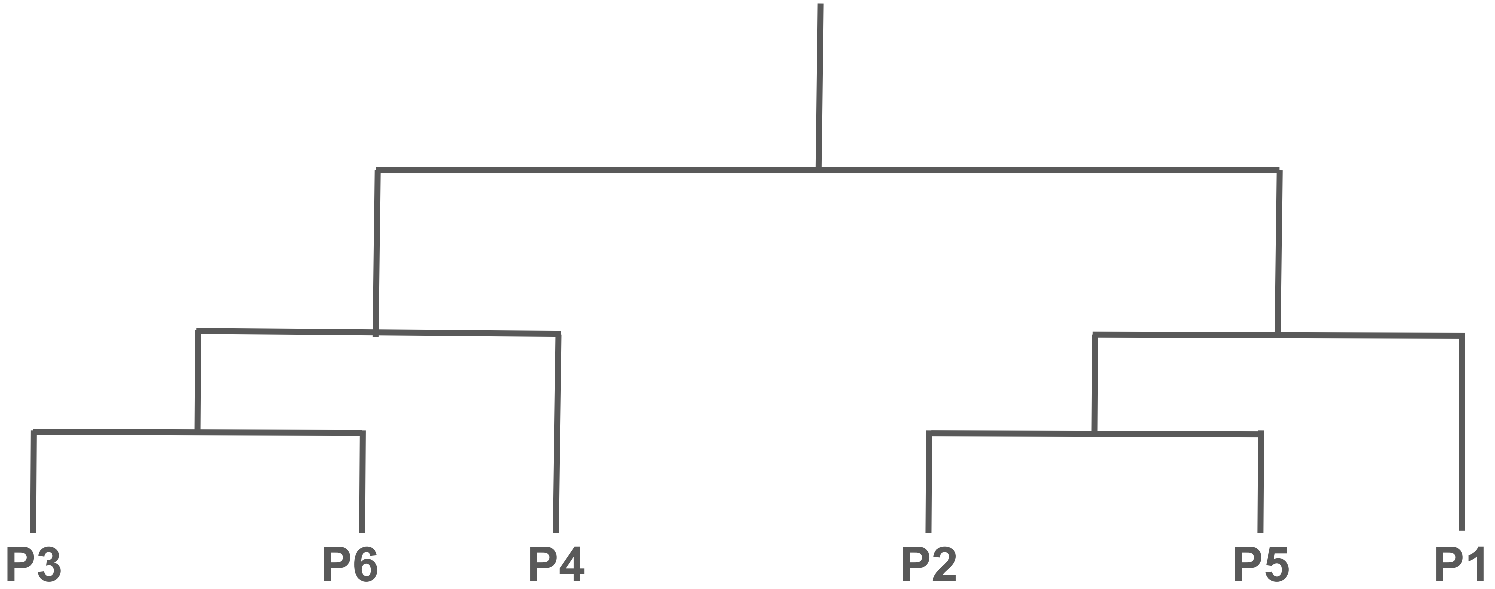
	P1	P2	P3, P6	P4	P5
P1	0				
P2		0			
P3, P6			0		
P4				0	
P5					0

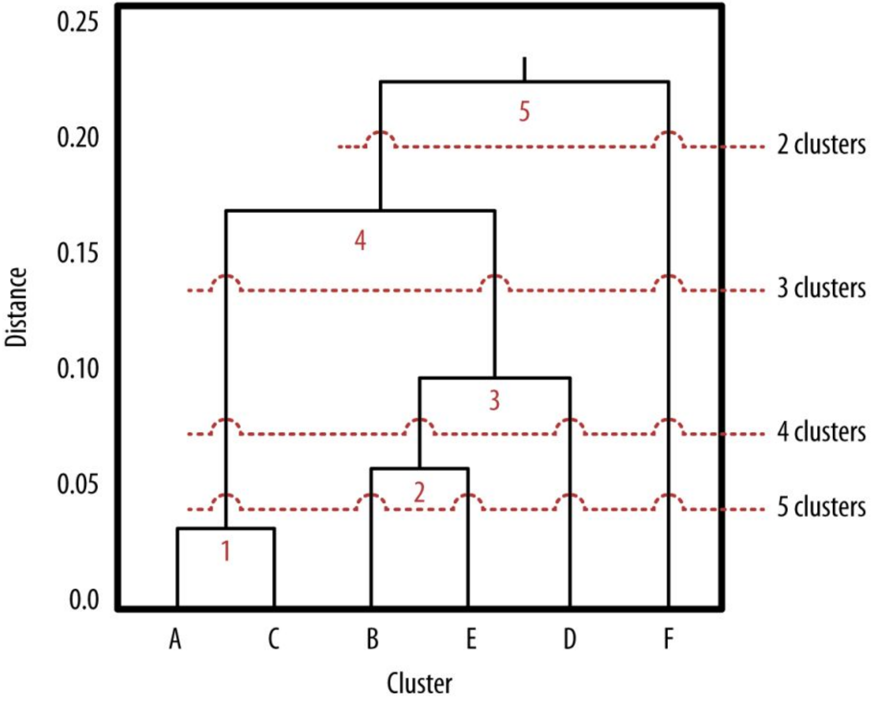
MAX((P3,P6), P1) = MAX(**(P3,P1)**, **(P6,P1)**) = MAX(0.22, 0.24) = **0.24**

	P1	P2	P3, P6	P4	P5
P1	0				
P2	0.23	0			
P3, P6	0.24	0.25	0		
P4	0.37	0.19	0.22	0	
P5	0.34	0.14	0.39	0.28	0


P3
P6
P2
P5

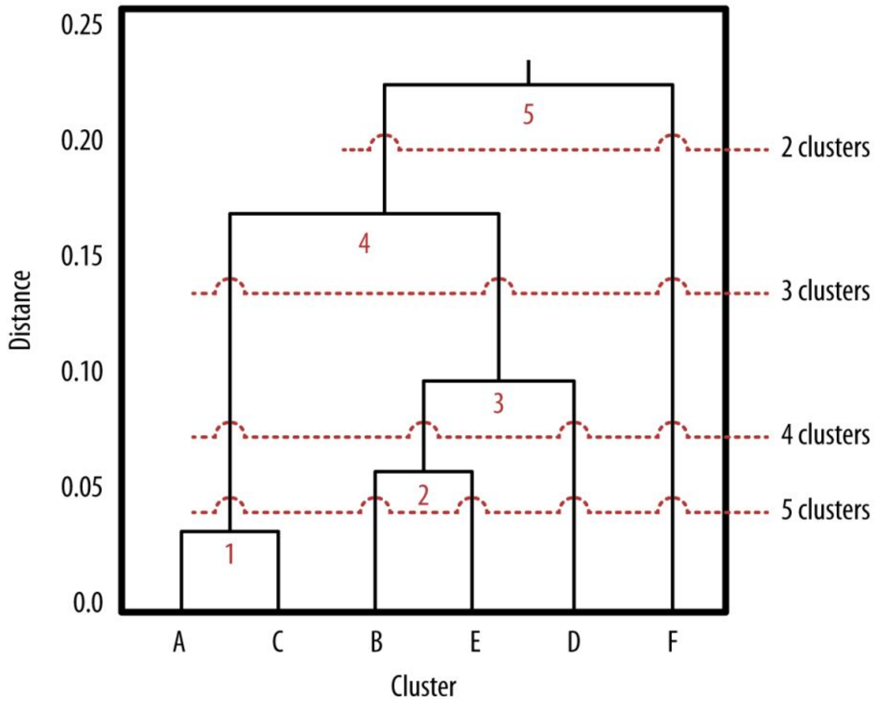
	P1	P2, P5	P3, P6	P4
P1	0			
P2, P5		0		
P3, P6			0	
P4				0


P3
P6
P4
P2
P5
P1

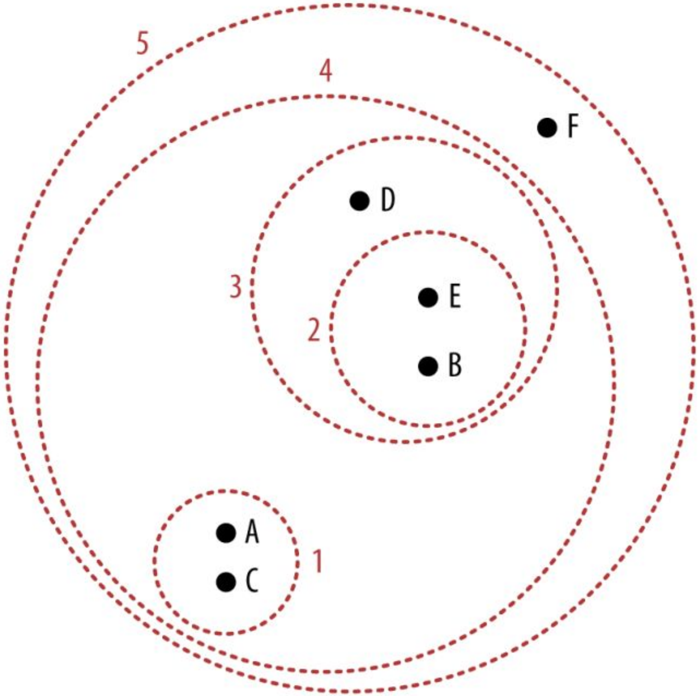


Nearest neighbor pairs are grouped to clusters

- A and C are closest so they are grouped first
- Followed by B and E
- The diagram is known as dendrogram



Dendrogram



Nested Clusters

In Summary

- Unsupervised-learning
- Clustering Concept
  - Similarity Measures
  - Distance Functions
  - Quality of Clustering
- Clustering Approaches
  - Partitioning-based
  - Hierarchical-based
  - Density-based

Ref: https://www.pinatafarm.com/p/4c995635-9593-40c9-9f0f-3b0c3ff5b8c5

Q & A